



DEPARTMENT OF INFORMATION TECHNOLOGY
LAB MANUAL

CS23432 – Software Construction

(REGULATION 2023)

RAJALAKSHMI ENGINEERING COLLEGE

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Year / Branch / Section: *II / IT / B*

Semester: *IV Sem*

Academic Year: *2025 - 2026*



RAJALAKSHMI ENGINEERING
COLLEGE
An Autonomous Institution

BONAFIDE CERTIFICATE

Name: HARINI, G.

Academic Year: 2025-26 Semester: IV Branch: Information Technology

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Certified that this is the bonafide record of work done by the above student in
the Software Construction CS23P32 Laboratory

during the academic year 2025-2026

Signature of Faculty-in-charge

Submitted for the Practical Examination held on 12.05.2026

Internal Examiner

External Examiner

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Azure Devops Environment Setup

Aim

To set up & access the Azure Devops environment an organization through the Azure portal.

Installation:

1. Open your web browser and go to the Azure website:

<https://azure.microsoft.com/en-us/get-started/azure-portal>.

Sign in using your Microsoft account credentials. If you don't have a Microsoft account, you can create one here:

<https://signup.live.com/?lic=1>

Get Started with Azure

Sign in

2. Azure home page

Azure services - Azure Devops Organizations.

3. Open Devops environment in the Azure platform by typing Azure Devops Organizations in the search bar.

4. Click on the My Azure Devops Organization link and create an organization & you should be taken to the Azure Devops Organization Home Page.

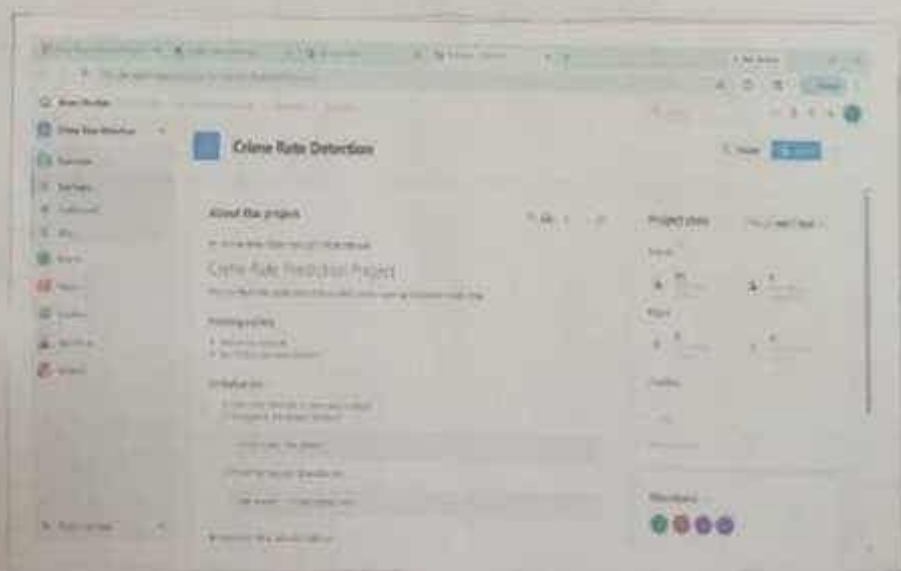
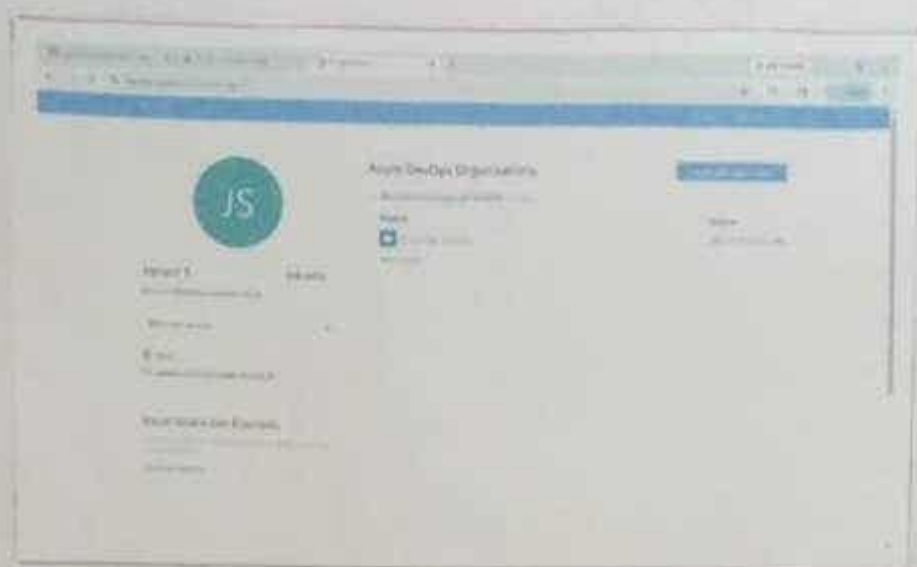
Azure DevOps

My Azure DevOps Organization



Result:-

✓ Successfully accessed the Azure DevOps environment and created a new Organization through the Azure portal.



Ex. No: 2

12.02.26

Azure DevOps Project Setup and User Story Management

Aim:

To get up an Azure DevOps project for efficient collaboration & agile Work management.

Procedure:

1. Create an Azure Account

Azure DevOps

harini.g.2024.it@rajalakshmi.edu.in

Name your Azure DevOps Organization

dev.azure.com / harini.g.2024.it@rajalakshmi.edu.in

2. Create the First Project in Your Organization

(a) After the Organization is set up, you'll need to create 1st project. This is where you'll begin to manage code, pipelines, work items & more.

(b) On the Organization's Home page, click on the new project button.

(c) Enter the project name, description, & visibility options.

name: Choose a name for the project.

description: Optionally, add a description

provide more context about the project.

Visibility: choose whether you want the project to be Private or Public.

d) once you've filled out the details, click Create to set up your 1st project.

Create new project.

Project name: Crime Rate Detection

Visibility: Public

Version control: Git

Work item process: harini.2024.it Agile

Create

3. Once you ~~logged~~ in, ensure you are in the correct ~~Organization~~. If you're part of multiple Organizations, you can switch ~~to~~ them. Click on Organization name, & you should taken to the Azure devops Organization Home Page.

(HG)

Harini G

harini.g.2024.it@rajalakshmi.edu.in

dev.azure.com/harini.2024.it (owner)

Project ☒ Crime Rate Detection

4. Project dashboard.

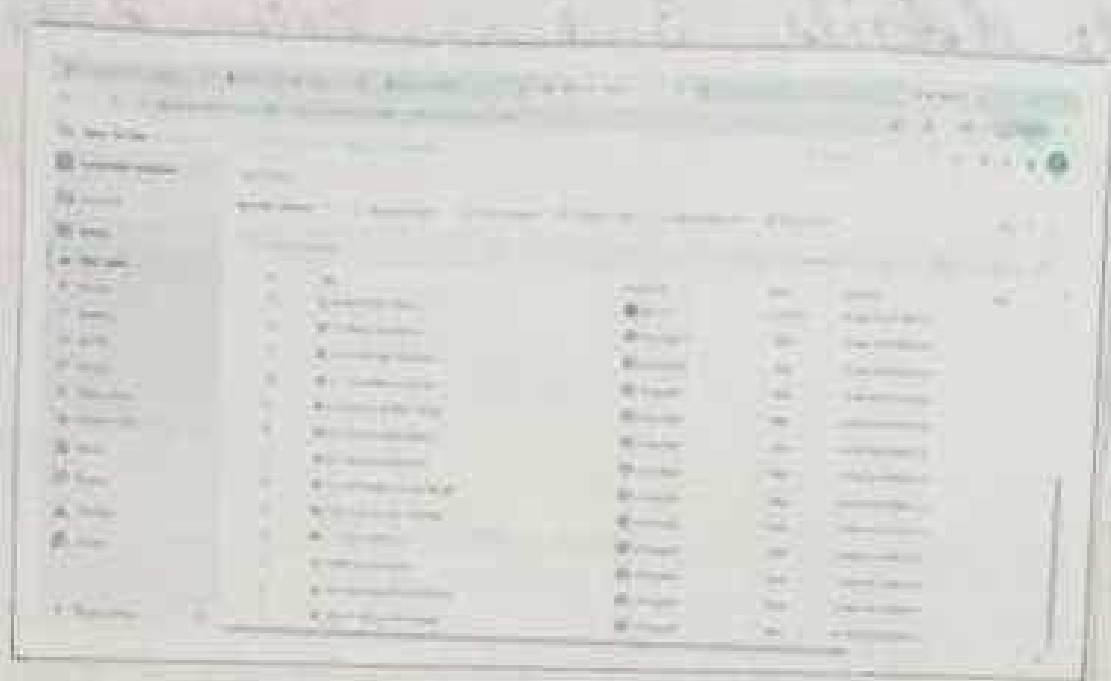
About the Project:

The Crime Rate Detection Application is a User-friendly web-based system designed to analyze & display crime data for different locations. It helps users understand crime trends in a particular area.

The application uses data sources or APIs to collect crime-related info. and presents details such as crime type, frequency, & location. It also includes features like location-based search, data visualization, & basic error handling, with a simple & easy-to-use interface.

Result:

Successfully created a Crime Rate Detection project with basic data analysis & user-friendly features.



Ex.no: 3

19.02.26

Setting up Epic, Features & User stories for Project Planning.

Aim:

To learn about how to create epic, user story, features, backlogs for your assigned Project.

Procedure:

Create epic, Features, User stories, Task

As a user, I want to view crime data & trends in different locations so that I can stay aware & take necessary precautions.

1. Fill in Epic.

Epic 1:

Crime Data Detection/Management.

2. Fill in Features

1. City / Area Search Feature
2. Crime Rate Display
3. Crime Rate Analysis
4. Location Based Crime Detection
5. Data Visualization (charts / graphs)
6. User Interface & Experience

3. Fill in User Story Detail

7. View crime data by city/area name
8. View types of crimes in a location
9. View crime rate trends over time
10. Detect user location automatically
11. Handle invalid location input
12. Display user-friendly interface

User Stories:-

1. As a user, I want to search crime data by entering a city or area name so that I can know the crime details of that location
2. As a user, I want to view different types of crimes so that I can understand the nature of crimes in that area.
3. As a user, I want to see crime rate trends so that I can analyze whether crime is increasing or decreasing.

Result:-

Thus the creation of epic, features, user story & task has been created successfully.

Ex. No: 4

26.02.26

Sprint Planning.

Ans:-

To assign user story to Specific Sprint for the Crime Rate Detection project.

Sprint Planning.

Sprint 1:

Task

1. View crime data by city/area

Assigned to: Harini

Iteration 1:

2. View types of crimes in a location

Assigned to: Harini

Iteration 2:

3. View crime rate trends (daily/monthly/Yearly)

Assigned to: Jayasree

4. Search location using input field.

Assigned to: Jayasree

Iteration 2:

5. Detect User location automatically
Assigned to : Nandhana
6. Handle invalid location input
Assigned to: Nandhana

Sprint 2:

7. Display crime data Using charts/graphs
Assigned to : Juvani
I1
8. Show crime hotspots on map
Assigned to: Harini

I2:

9. Alerts & notifications for high crime areas
Assigned to: Jayasree
10. Display user-friendly interface
Assigned to: Team Member 1.

Result:

The Sprints are Successfully created for the Crime Rate Detection App project.

Ex. No: 5

08.03.26

Project Estimation

Aim:

To create a proper estimation for the user stories of the Crime Rate Detection Project.

Project Estimation:

Story Points Measure:

Effort

Complexity

Time

Using Fibonacci Scale: 1, 2, 3, 5, 8, 13...

Story Points for Crime Rate Detection App:

User Story	Story points	Reason
Search crime data by city	3	API Integration + Input Handling
View crime details of selected location	2	Simple data Display
View diff types of crimes	3	Data Categorization

View crime
date trends

5

Data processing +
UI charts

Detect User
location
automatically

5

GPS + permission
handling

Handle Invalid
location i/p

2

Error handling

Show user-friendly
Interface

3

UI design &
styling

Total = 23 story points

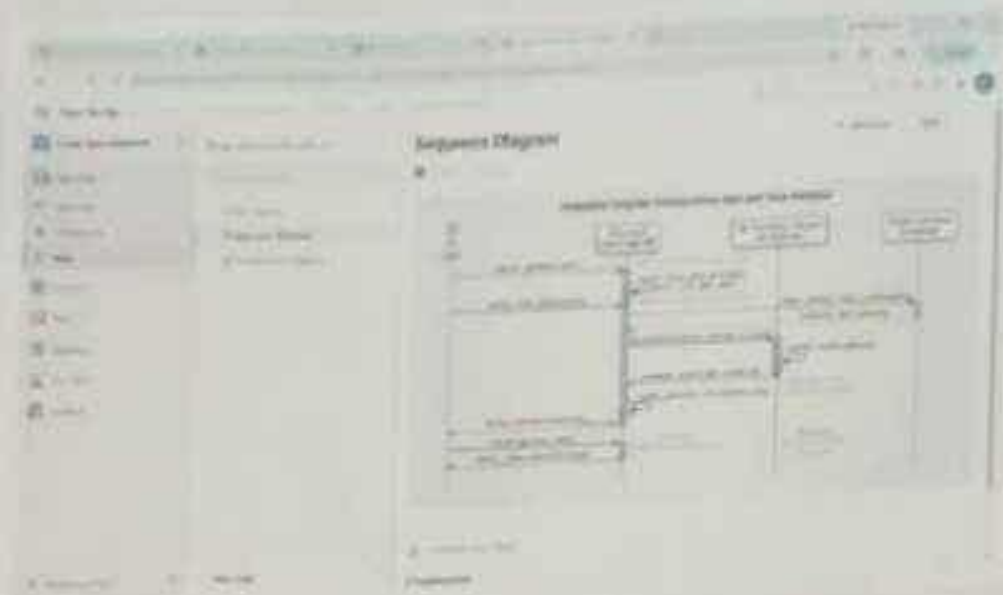
Steps:

Boards → Work Item → User Story →

Assign Story Points → (1/2/3/5/8) → Save

Result:

The estimation is successfully created
for the Crime Rate Detection project using
Project Estimation.



Exp no: 6

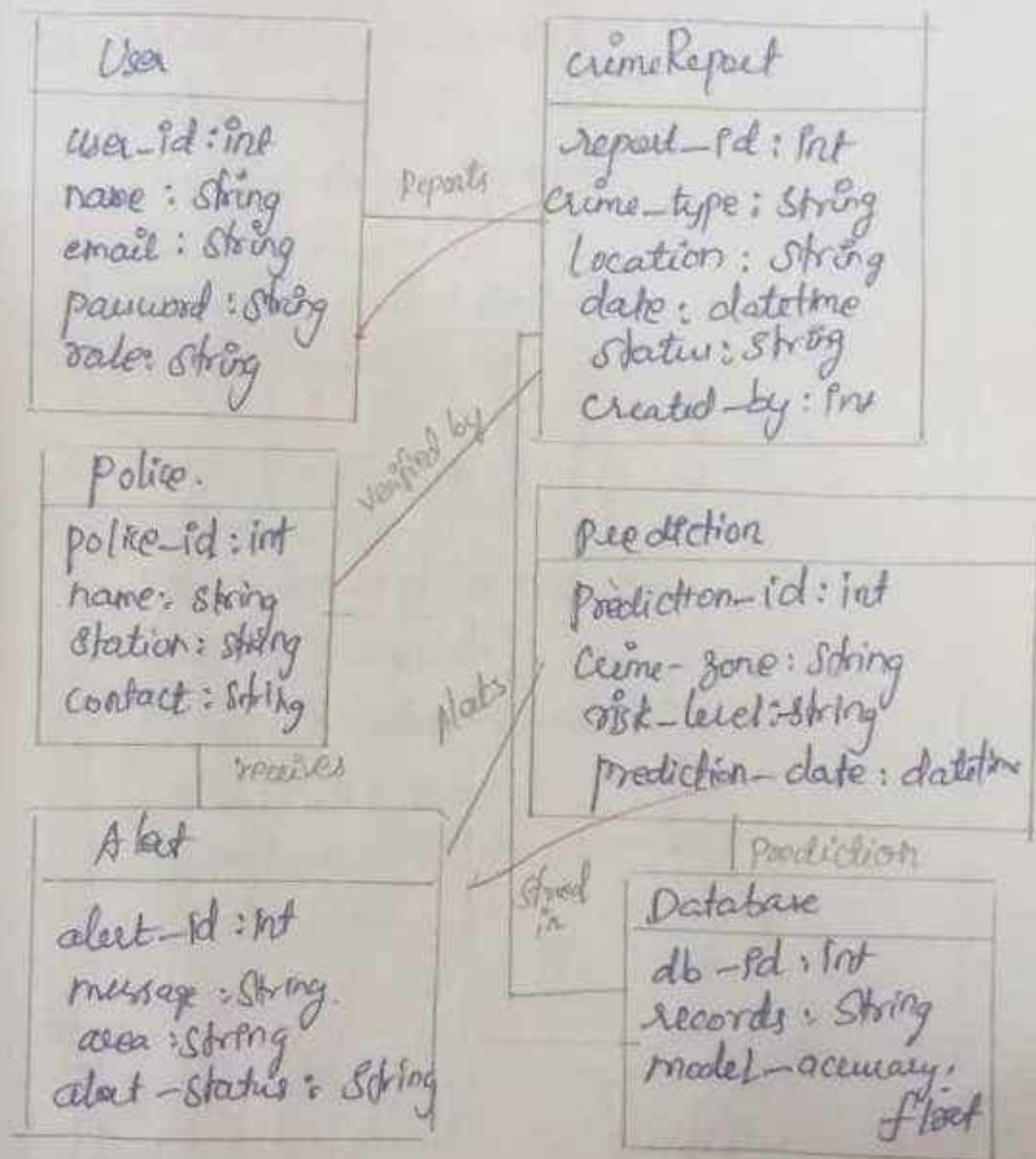
Date: 17/3/26

Designing class & sequence diagrams for Project Architecture.

AIM:

To design a class diagram and sequence diagram for the given project.

Class Diagram



6B. Sequence Diagram

Actors:

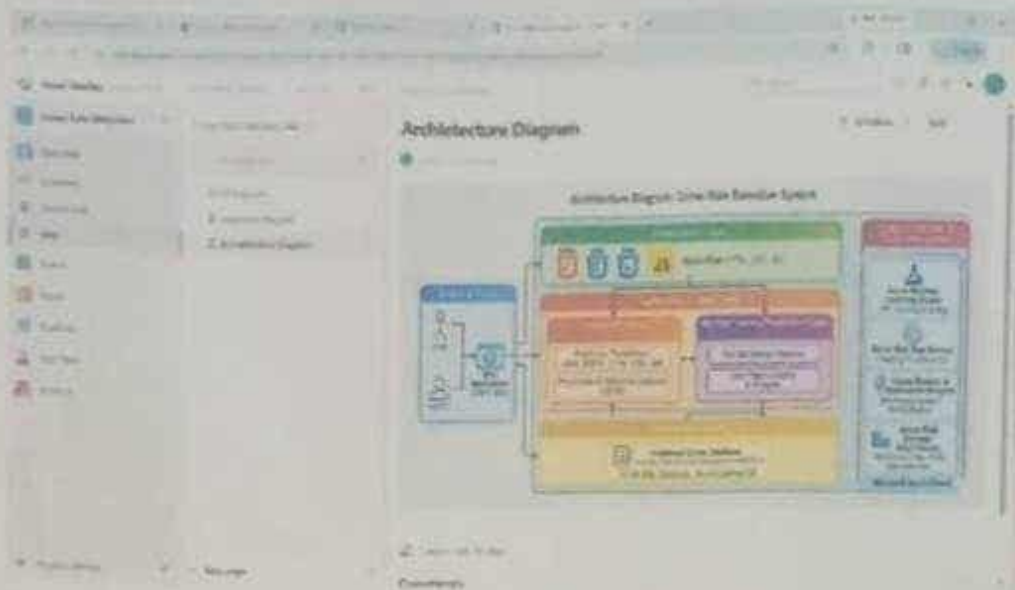
User
Mobile / web App
Crime Controller
Database

Flow:

1. User → Mobile App
Login (email, password)
2. Mobile App → crime Controller
Send Login Request
3. Crime Controller → Database
Verify User
4. Database → crime Controller
Login Success.
5. User → Mobile App
Report Crime
6. Mobile App → crime Controller
Fetch Crime data
7. Crime Controller → Database
Store Crime Data
8. Database → crime Controller
Data Saved

Result:

The class diagram & sequence diagram are designed successfully for Crime Rate detection System.



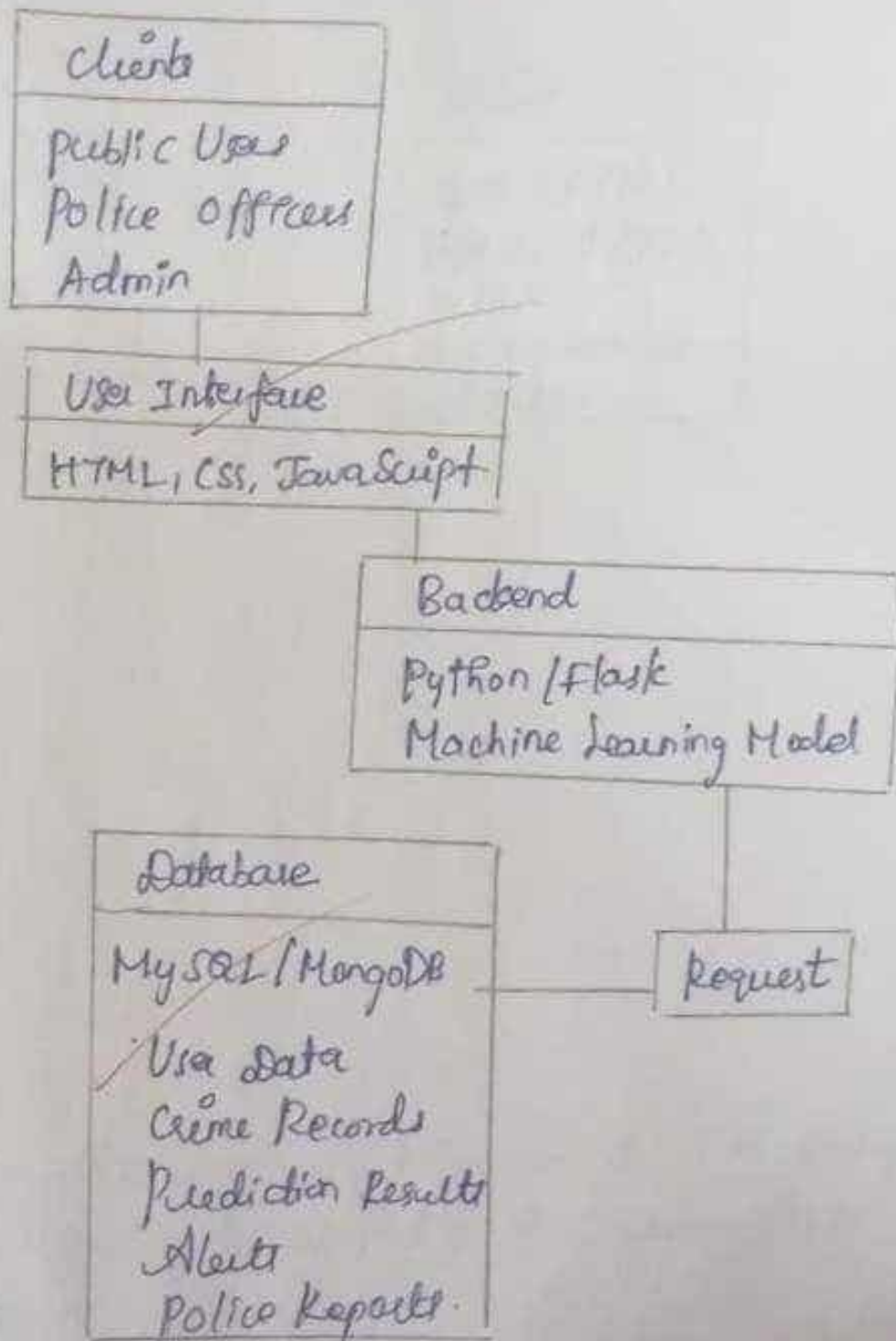
Ex. NO: 7
2/04/26

Designing Architectural & ER diagram for Project Structure

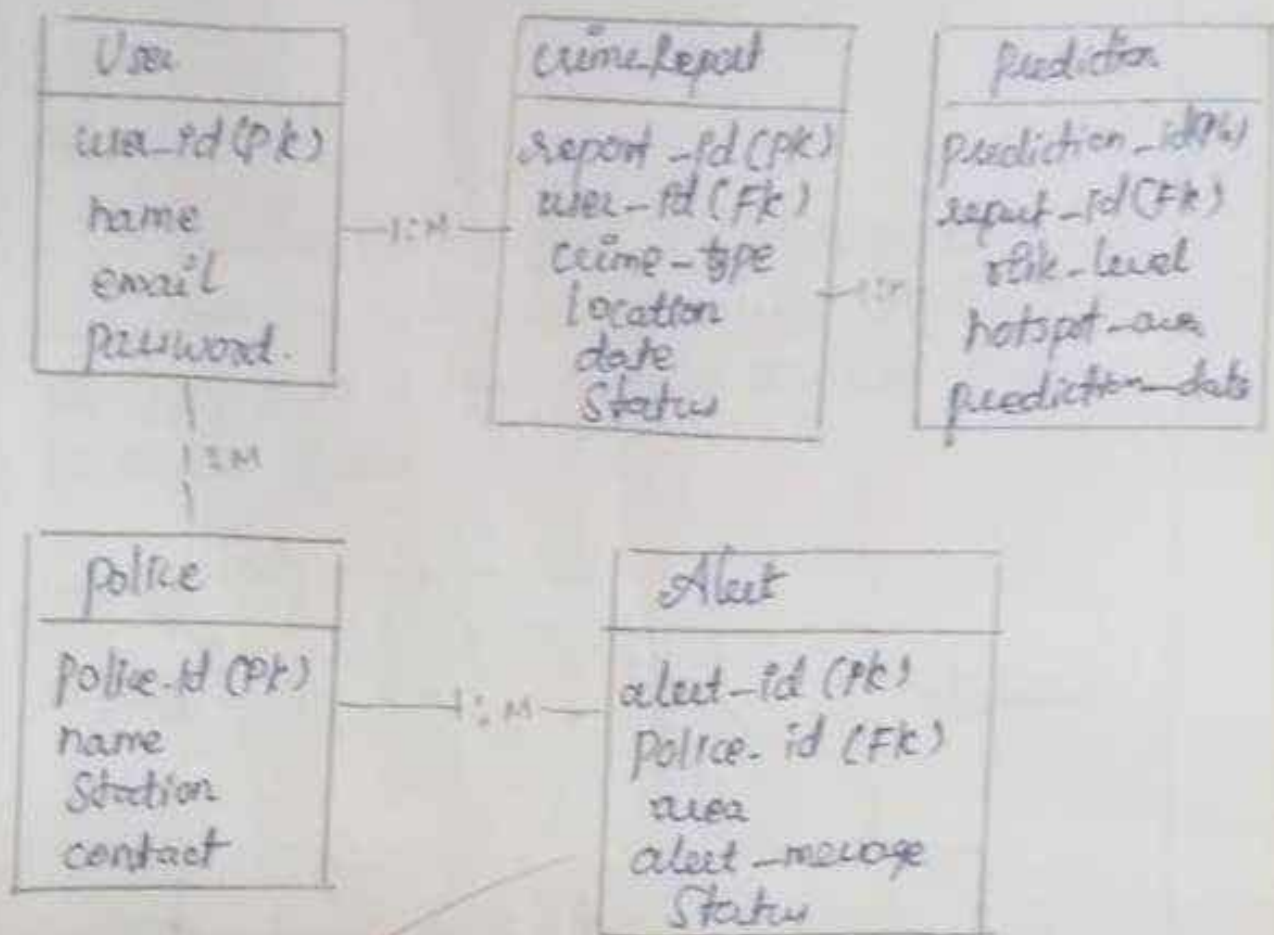
AIM:

To design an Architectural Diagram & ER diagram
for Crime Rate Detection System.

TA - Architectural Diagram



78. ER Diagram



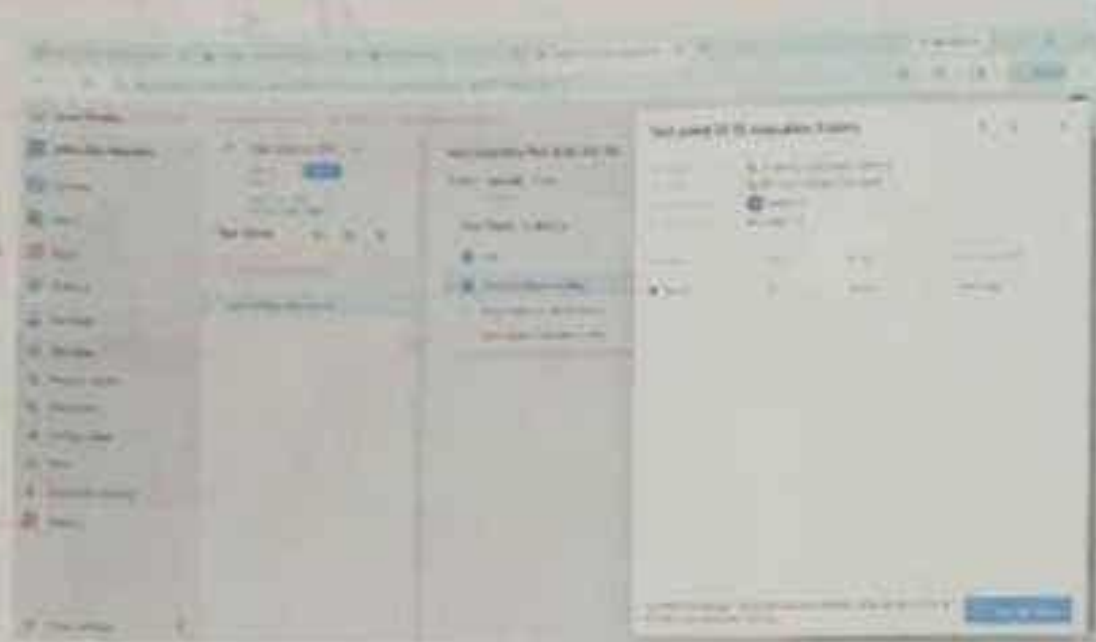
Result :-

The Architecture Diagram & ER Diagram is designed successfully for Crime Rate Detection System.

Handwritten notes on the left side of the page, including the word "Handwritten" and other illegible text.

Handwritten notes in the top center of the page, including the word "Handwritten" and other illegible text.

Handwritten notes on the top right side of the page, including the word "Handwritten" and other illegible text.



Handwritten notes on the right side of the page, including the word "Handwritten" and other illegible text.

Ex. No: 08

9/4/26

Testing - Test Plans & Test Cases

AIM:

To design test plans & test case & create test case for at least five user stories showcasing the happy path & error scenarios in Crime Rate Detection System.

Test Case Design Procedure:

1) Understand Core Features of the Application

User Registration & login

Report Crime

Crime Prediction (ML model)

View Crime Hotspots

Police Mgmt.

2. Define User Interactions

Logging in

Reporting a Crime

Viewing predictions

Checking alerts

Admin managing records.

3. Design Happy path test Cases

Validates System works under normal

Conditions.

4. Design Error path Test Cases

Tests invalid or unexpected scenarios

eg:-

login fails with wrong password

5. Break Down steps & Expected results.

Each test case includes:

Step by-Step actions

Expected op.

6. OR clear Naming & IDs

TC01 - Successful login

TC02 - login failure

Test Case

TC01 Successful Sign Up

Action:-

Go to Sign-up page

Enter valid details

click "Sign Up"

Result:-

Account created successfully

Redirected to dashboard

Type: Happy Path.

TC02 - Successful Login

Action:

- Go to Login Page
- Enter valid credentials
- Click "Login"

Expected Result:

- Login Successful
- Dashboard displayed

TC03 - Login failure

Action:-

- Enter invalid password
- Click "login"

Result:

Error message displayed

Type: Error path.

User Stories:

- As a user, I want to register & log in so that I can access the system.
- As a user, I want to report crimes so that authorities can take action.
- As a police officer, I want to monitor crime.

Result:

Thus, the test plan and test cases for the Crime Rate detection System are created successfully.

Ex no: 2

16/04/26

Load Testing & Detection

Aim:

Aim is to perform load Testing & implement pipeline for the crime rate detection.

Load Testing.

Tests system performance under multiple users & identifies delays or issues.

Step to create Azure load Testing:

1. Sign in to Azure Portal.
2. Search & click create (Azure load)
3. Fill : Subscription, resource name, region
4. Configure tags & click review & create.
5. After deployment, go to resource.

Pipelines:-

Steps:-

1. Connect Github repository to Azure

DevOps

2. Create a new pipeline & select Github as a source
3. create azure-pipeline.yml in repository

Run & Monitor pipeline.

- commit changes to trigger pipeline
- View logs in Azure DevOps
- Check for successful execution or error.

Result:-

Successfully performed load testing & demonstrated pipeline execution for the Crime Rate detection Project.

Ex. no: 10
23/4/26.

Project Structure & Naming Conventions.

AIM:

To provide a clear & organized view of the project folder structure & file naming conventions, helping contributors & users easily understand, navigate & extend the crime rate detection. Project. It ensures proper management of datasets, models, scripts, & output files.

GitHub Project:

<https://github.com/yourusername/crime-rate-detection>

Result:

The GitHub repo clearly displays the organized project structure & consistent naming conventions, making it easy for users & contributors to understand, maintain & navigate the crime rate detection.